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Task Description:

Historically, the study of polymeric materials has been characterized by the prevalence of shear flow over elongational flow experiments. This disparity might be attributed to the relative difficulty of conducting elongational experiments rather than to the lack of relevance in the polymer community. In fact, elongational flows have many applications in industrial processes, as well as in theoretical and numerical research, as they provide a discriminating test for models and theories. In this regard, the data provided by the SHERE (Shear History Extensional Rheology Experiment), SHERE II, and SHERE-R experiments offer a unique advantage by eliminating two of the main difficulties encountered in Earthbound elongational flow experiments: gravity and the inability to observe the evolution of the flow at long time scales.
The capillary thinning experiments, performed in the International Space Station, consist of stretching a piece of fluid placed between two disks and letting surface tension drive the fluid motion. The SHERE flight experiments further investigate and quantify, for the first time, the transient extensional stress growth characteristics of a dilute polymer solution after a defined pre-shear history. The SHERE data revealed that the extensional viscosity tends to decrease with an increasing degree of pre-shear for fixed extension rate. Similarly, a general trend that was observable was a systematic reduction in extensional viscosity with increasing extension rate for fixed pre-shear conditions. This is an effect that has not been reported in filament stretching experiments performed in 1-g conditions. This information is particularly relevant for understanding the deformation and evolution of non-Newtonian thin fluid columns in a wide variety of industries, including fiber-spinning, injection molding, or food and consumer product processing.

One of the main challenges in the modeling of viscoelastic materials is the selection of an appropriate constitutive model. Unlike the case with simple Newtonian fluids, there is not a "universal" constitutive equation capable of describing the flow of all viscoelastic material. In contrast, we are faced with the task of determining which equation(s) describes best a given material and/or process. Finding the appropriate model for a given material is a critical step in building truly predictive modeling platforms that will allow us to simulate conditions that are hard to access experimentally. The validity of constitutive theories for polymeric materials has not yet been confirmed due to the lack of experimental data obtained in extensional flows. In order to take advantage of the unique SHERE data, this work will present the first-ever detailed analysis and numerical simulation of filament stretching and capillary thinning with a predefined shear history of models specifically developed to describe dilute polymer solutions. As a first look, a 1+1-D Lagrangian formulation of the models will be used, similar to earlier work on filament stretching of wormlike micelles performed by the principal investigator. The proposed projects investigate the effects of a pre-shear history on the transient extensional viscosity in a filament stretching flow for a viscoelastic material using data obtained in the International Space Station during the SHERE, SHERE II, and SHERE-R projects. Ultimately, the SHERE experimental data will provide a testing ground for the viability and accuracy of existing polymer constitutive theories and/or provide insight into new directions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The SHERE flight experiment investigates and quantifies, for the first time, the transient extensional stress growth characteristics of a dilute polymer solution after a defined pre-shear history. This information is particularly relevant for understanding the deformation and evolution of non-Newtonian thin fluid columns in a wide variety of industries, including fiber-spinning, injection molding, or food and consumer product processing. This research will describe the dynamic network that results from the complex interactions among microscopic constituents and how such interactions dictate material properties and functions at the macro-scale.

Task Progress:

The Shear History Extensional Rheology Experiment (SHERE) flight experiment investigates and quantifies, for the first time, the transient extensional stress growth characteristics of a dilute polymer solution after a defined preshear history. This information is particularly relevant for understanding the deformation and evolution of non-Newtonian thin fluid columns in a wide variety of industries, including fiber-spinning, injection molding, or food and consumer product processing. Constitutive equations for viscoelastic fluids such as dilute polymer solutions generally predict large changes in the extensional rheology of the material as the elongation rate is increased. However, the validity of these theories has not yet been confirmed due to the lack of experimental data obtained in extensional flows.
In order to take advantage of the unique SHERE data, this work focuses on the development of the first-ever detailed analysis and numerical simulation of capillary thinning of models specifically developed to describe dilute polymer solutions. The proposed projects investigate the effects of a pre-shear history on the transient extensional viscosity in a filament stretching flow for a viscoelastic material using data obtained in the International Space Station (ISS) during the SHERE, SHERE II, and SHERE-R projects. Ultimately, the SHERE experimental data will provide a testing ground for the viability and accuracy of existing polymer constitutive theories and/or provide insight into new directions. Our approach combines microscopic structure and dynamics, consistent with statistical physics of dilute polymer solutions, with 1) Earthbound experiments to guide the linear viscoelastic mode section, and 2) simulations to fully couple microstructure dynamics and macroscopic deformation for nonlinear characterization. We have considered the surface tension driven thinning of a polymeric liquid filament. Using slender jet theory, we simulated the thinning of filaments governed by macro homogeneous and inhomogeneous constitutive models, and multiscale simulations where polymers are governed by Langevin equations.

For the macro simulations, we assume periodic boundary conditions. The velocity and polymer stress are assumed to be initially uniform, while an initial defect is imposed on the radius. The nonlinear system of partial differential equations is solved numerically using the method lines. All spatial terms, except advection, are discretized by second-order central differencing. Advection terms are treated by a scheme that is weighted locally via the velocity. In particular, the derivative is interpolated between pure central differencing and pure upwind as the magnitude of the velocity increases; in regions of slow flow, a central differencing is used, however, as the velocity increases, the weighting moves closer to upwind differencing. We employ the implicit Euler method for time discretization, which we have further enhanced by implementing time filters that increase the accuracy from first to second order. Additionally, we have implemented variable step size, which improves the speed and efficiency of the solver. This solver has been benchmarked against existing capillary thinning results.

For the homogeneous models, we considered the FENE-P and Giesekus models in order to extend the parameter space for the formation of satellite beads. In particular, we simulated the effects of finite extensibility and anisotropic drag. We ultimately showed that polymer stretch is directly connected to filament evolution. Satellite bead formation can be prevented with higher degrees of polymer stretch. This stretch can be increased by either increasing the extensibility or by decreasing the drag.

Under inhomogeneous flow conditions, standard diffusive process acts to suppress concentration gradients by moving polymer down concentration gradients, i.e., polymers move from high to low concentrations. On the other hand, polymer molecules are pulled towards regions of higher stress. This term produces a similar mechanism observed in two-fluid models for polymer solutions. In particular, the theoretical foundation of the Helfand-Fredrickson (HF) mechanism for concentration fluctuation amplification is the coupling in a polymer solution between polymer concentration and polymer transport properties: specifically, regions of higher concentration will exhibit higher viscosity and elastic modulus. Under flow, these couplings may result in a local concentration inhomogeneity producing a polymer stress gradient that drives polymer up the concentration gradient, in opposition to diffusion. To capture this effect, we considered a model that described an ensemble of non-interacting Hookean dumbbells whose configuration varies both on space and time, which allows for an explicit formulation of molecular migration while considering the polymer molecules as extended objects in space. Simulations showed that elastic stress gradients are largest at the interface between beads and string. Thus, via the HF mechanism, we predict a buildup of polymer at the pinchoff point connecting bead and string. This enhanced concentration should inhibit filament breakup.

The constitutive equations presented above are continuum level equations that have been averaged from the mesoscopic level; in particular, the conformation tensor is the second moment of the number density distribution function. In order to be more faithful to physics, we can instead solve directly for the configuration at each point in space. In this approach, the solution of the equations is carried out through a multiscale simulation scheme: i) the macroscopic flow and distribution in space of number densities are solved using the conservation of mass and momentum equations; ii) the microscopic evolution of the network is evolved by stochastic differential equations describing dumbbell configurations. The Langevin equations describing the behavior of an elastic dumbbell are solved on GPUs, which allow for massively parallel simulations. Utilizing the dumbbell equations directly provides for more accurate information on the stretch and orientation of polymer chains than constitutive equations arising from solving for the second moment of the end-to-end vector. For example, the stretching and orientation of dumbbells are captured in a similar manner in which an experimentalist will analyze fluid behavior using scattering techniques.

Under capillary thinning, we investigated the effects of the Deborah number, wavenumber, and maximum extensibility on the formation of satellite drops. As the elasticity of the fluid is increased, the formation of satellite droplets is inhibited. This inhibition is related to a higher tensile stress in the filament prior to the formation of the droplets, represented by a larger expansion ratio. This in turn hinders the reorganization of the strands that would result in a lower expansion ratio in the middle of the jet, which is what at the end leads to the formation of droplets.

Bibliography Type:	Description: (Last Updated: 03/04/2024)
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